RENDY RYAN RENALDI

Engineering Physics Graduate | Instrumentation, Calibration & Equipment Monitoring

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PROFILE SUMMARY

Engineering Physics graduate from Telkom University with hands-on experience in sensor integration and calibration, electrical circuit testing, troubleshooting and technical documentation through 2 engineering internships and 3 laboratory teaching roles. Project experience includes centrifugal-pump RPM monitoring, calibration-chamber development with a Mass Flow Controller, low-voltage wiring and power-distribution design, and the assembly and testing of 7 wearable sensor units. Experienced in comparing measurements against reference instruments, investigating inconsistent readings and documenting test results. Coordinated 11 laboratory assistants supporting over 100 students. Supported by foundations in industrial automation, thermodynamics, fluid mechanics, heat transfer, reliability and safety, seeking an entry-level role in instrumentation, equipment monitoring, maintenance support or technical operations in mining, energy or process industries.

EDUCATION

Bachelor of Engineering — Engineering Physics *Sep 2022 – Jul 2026*

Telkom University — GPA 3.37/4.00; study programme accredited Unggul

- Graduated on 23 July 2026; completed all coursework and the final-project defence, with an official graduation statement issued.
- Relevant coursework: Electrical Circuits and Electronics, Measurement Systems, Instrumentation Systems, Engineering Thermodynamics, Fluid Mechanics, Thermal Analysis, Heat Transfer, Industrial Automation, and Reliability and Safety.
- Final project: an IoT wearable air-quality monitoring system integrating Human Activity Recognition (HAR).

WORK AND INTERNSHIP EXPERIENCE

Measurement and Instrumentation Laboratory, Telkom University — Bandung

Head of Laboratory Assistants (Sep 2025 – Jul 2026) · Part-time, 2–3 shifts/week

- Promoted to lead 11 assistants supporting more than 100 students after previously serving as a teaching assistant in the same laboratory.
- Supported Instrumentation Systems circuit testing using DC power supplies, oscilloscopes and function generators; assisted signal checks for filters, voltage–current conversion, ADC/DAC, instrumentation amplifiers and Arduino Uno.
- Supported Measurement Systems practicals covering data analysis and position, angle, temperature, flow and light sensors.
- Standardised laboratory data collection and reporting, organised schedules/task allocation and monitored activity delivery and team coordination.

PUI-PT Intelligent Sensing-IoT, Telkom University — Bandung

Research Intern — MBKM Programme (Sep 2025 – Dec 2025) · Part-time alongside coursework

- Completed a semester-long research internship supporting two PUI-PT IS-IoT products: TrapGradien (ioniser-based air filtration) and AirGradien (wearable air-quality monitoring).
- Selected/procured sensors, assembled and tested sensor–microcontroller connections, and built/improved a calibration chamber for system evaluation.
- Disassembled an AirGradien unit to examine sensor positioning and airflow; studied chamber MFC operation, then reassembled and integrated sensors, battery power and website connectivity.
- Installed multiplexers for parallel calibration of matching sensor types and prepared a 3D printer for TrapGradien and AirGradien enclosures/fixtures.
- Prepared a final report documenting sensor selection through integration, reviewed and approved by academic and PUI-PT IS-IoT field supervisors.

PT SAS Aero Sishan — Bandung

Systems Division Intern — MATS Project (Jun 2025 – Aug 2025) · Full-time

- Completed a Systems Division internship on MATS, a defence prototype collaboration involving PT SAS Aero Sishan, Telkom University and Pussenkav TNI AD; the project underwent testing/demonstration for the military.
- Designed Pixhawk 6C ROV wiring architecture, considering component voltages, signal flow and port allocation; reviewed designs in weekly technical meetings with partners.
- Modified the telemetry interface using Qt Creator and QGroundControl, supported by Visual Studio, to add front/rear camera feeds to an ROV interface mockup.
- Designed 3D-printable prototype casings and electronics enclosures in Autodesk Inventor for Pixhawk 6C, batteries and other components; revised designs following team reviews.
- Used a Waterfall/V-Model approach and delivered ROV power-distribution schematics and a GCS mockup presented to the client as a basis for fabrication.

Instrumentation System Laboratory, Telkom University — Bandung

Teaching Assistant — Electrical Circuits and Electronics (Jun 2024 – Jul 2025) · Part-time, 2–3 shifts/week

- Taught series/parallel circuits, Thevenin/Norton, node/mesh analysis and passive filters; assisted circuit troubleshooting and measurement interpretation using DC power supplies, oscilloscopes and function generators.
- Taught Basic Electronics practicals covering diode, BJT, MOSFET and op-amp circuits.
- Taught Digital Electronics: logic gates, Boolean/Karnaugh maps, encoders/decoders, multiplexers/demultiplexers, seven-segment displays, flip-flops, multivibrators and counters.
- Served on the Practicum Division, revising modules and preparing laboratory materials.

Instrumentation System Laboratory, Telkom University — Bandung

Assistant Recruitment Committee Chair — SILO (Feb 2025 – Jun 2025) Part-time

- Led assistant recruitment through document screening, written tests, instrument tests, teaching demonstrations and interviews.
- Coordinated assessment of 24 applicants, resulting in recruitment of 11 assistants through multiple selection stages.

Basic Computing Laboratory, Telkom University — Bandung

Teaching Assistant — C Programming (Jun 2023 – Jun 2024) · Part-time, 3–4 shifts/week

- Taught C programming fundamentals and debugging to more than 200 students in laboratory practicals; selected through the Kost-In C application project and served on the Rules and Discipline Committee.

RELEVANT SKILLS

- Instrumentation and calibration: sensor selection and integration; calibration-chamber setup; reference-instrument comparisons; temperature, humidity, flow, position and light measurement; evaluation of sensor response and measurement deviations.
- Electrical testing and troubleshooting: read and prepare circuit/wiring diagrams, check connections and component voltage requirements, observe signals and support fault diagnosis using oscilloscopes, DC power supplies and function generators.
- Equipment monitoring and data acquisition: centrifugal-pump RPM measurement using a TCRT5000 optical sensor; ESP32/ESP8266 sensor acquisition, signal conditioning, ADC/DAC, voltage–current conversion and remote monitoring through Firebase, Kodular and ThingSpeak.
- Automation and communication: Modbus RTU/RS-485 integration of a Mass Flow Controller in a calibration chamber; C/C++ microcontroller firmware; basic Omron CX-Programmer and PLC/HMI through academic practicals and simulation.
- Electrical and mechanical integration: low-voltage power-distribution design, battery and voltage-regulator evaluation, KiCad schematics/PCB design, Autodesk Inventor enclosures, prototype assembly and chamber airflow evaluation; academic thermodynamics, fluid mechanics, heat transfer, reliability and safety.
- Data analysis and documentation: Python and Microsoft Excel for timestamp alignment, outlier filtering, calibration analysis and visualisation; Microsoft Word for technical reports; standardised laboratory records and supervisor-reviewed engineering documentation.
- Coordination and communication: led 11 assistants supporting over 100 students, organised schedules and task allocation, coordinated recruitment, and delivered practical instruction; EPrT 570/677, CEFR B2 based on Telkom University mapping.

RELEVANT PROJECTS

Pump RPM Monitoring — Equipment Condition Monitoring *Sep 2025 – Jan 2026*

- Contributed to TCRT5000 optical-sensor integration for centrifugal-pump RPM monitoring in a team project with PT GLS and PT Trengginas Jaya; sent readings to Firebase every few seconds and created a live display in Kodular.
- The team installed and tested the system at Telkom University dormitory pump facilities to collect operating data for equipment-condition evaluation; contributed to machine-learning data processing.

TrapGradien — Sensor Integration and IoT Air Quality Monitoring *Sep 2025 – Dec 2025*

- Compared sensor specifications, operating principles, accuracy, power consumption and microcontroller compatibility; supported procurement and bench testing for an MBKM air-filtration research project.
- Designed a Wemos D1 mini sensor PCB in KiCad, assembled breadboard/perforated-board prototypes and tested sensor connections; tested Wi-Fi data transmission through PHP to a MySQL-backed dashboard using a supervisor-provided database.
- Improved a particulate calibration chamber, connected its Mass Flow Controller through Modbus RTU/RS-485 and tested airflow; installed multiplexers for parallel sensing and evaluated battery configurations and voltage regulators.

ROV — Electrical Schematics, Power Distribution and Enclosure *Jun 2025 – Aug 2025*

- Designed a Pixhawk 6C electrical block architecture and power-distribution documentation considering component voltages, signal paths and port allocation; planned GPS and front/rear camera telemetry.
- Modified the Qt Creator/QGroundControl interface to add camera feeds and designed casings/electronics enclosures in Autodesk Inventor; produced technical schematics and interface mockups for subsequent fabrication.

AirGradien — Sensor Assembly, Testing and Calibration *Feb 2026 – Jun 2026*

- Designed and built seven wearable sensor units using ESP32/ESP8266 with SPS30, ENS160 and AHT21 for particulate matter, TVOC, estimated eCO_2, temperature and humidity monitoring.
- Developed C/C++ firmware and Wi-Fi connectivity, checked sensor integration and compared readings with reference instruments; integrated Nicla Sense ME motion sensing on one unit.
- Processed calibration results and selected quadratic $PM_{2.5}/PM_{10}$ and linear TVOC, temperature and humidity corrections; documented interpretation limits for ENS160 eCO_2 as an estimate.

Sensor Testing — Calibration and Measurement Analysis *Feb 2026 – Jun 2026*

- Prepared chamber, outdoor rotation-platform and field-activity tests; compared two dynamic units with three static controls across four chamber experiments.
- Processed sensor/reference data in Python/Excel through timestamp alignment, physical-range checks, outlier filtering and inter-device comparisons.
- Documented test results in plots and reports; examined airflow, inlet orientation and device variation as possible causes of measurement deviations.

COMMUNITY SERVICE EXPERIENCE

Hardware and IoT Systems Developer — Smart Recovery Box *Aug 2024*

Arjasari Village, Bandung Regency — Telkom University Community Service

- Developed an IoT Smart Recovery Box with UV sterilisation, DHT22 temperature/humidity monitoring and ESP32 automatic control.
- Contributed hardware design, programming and system integration; supported installation and on-site troubleshooting at a poultry farm.

IoT Systems Team Member — Hydroponic/Aquaponic Greenhouse *2023*

Tarumajaya Village, Bandung Regency — Telkom University Community Service

- Contributed to implementation of an IoT hydroponic and aquaponic greenhouse to support community farming activities.

ACHIEVEMENTS

Top 10 Finalist — Klikstart Startup Competition *2025*

- Developed and presented the AirGradien wearable air-quality monitoring business concept, reaching the student-category Top 10 in the Bandung Techno Park competition.
- Prepared component-cost estimates, proposed selling prices, a business model and market analysis for the AirGradien proposal and presentation.

ORGANIZATIONAL EXPERIENCE

Supervisory Commission Chair — Engineering Physics Student Council *Aug 2025 – Jan 2026*

- Led oversight and performance evaluation of the student executive organisation.

Human Resources Staff — Engineering Service Community *Apr 2025 – Jan 2026*

- Supported recruitment, selection, onboarding and coordination of member-development programmes.

Cadre Development Department Staff — HMTF Unitel *Sep 2024 – May 2025*

- Organised member development and succession activities; chaired STATIS, the HMTF internship programme.

Mentor — Kalibrasi 2024 Organizing Committee *2024*

- Mentored 11 new members in leadership, confidence, solidarity and empathy.

Member — Student English Society *Sep 2022 – Sep 2023*

- Participated in English competitions and academic activities to develop communication and public-speaking skills.

CERTIFICATIONS

- English Proficiency Test (EPrT) — Score 570/677; CEFR B2 (Upper-Intermediate) based on Telkom University mapping, Telkom University Language Center (2026)
- Measurement Laboratory Coordinator & Teaching Assistant — Instrumentation System Laboratory, Telkom University (2026)
- MBKM Research Internship — PUI-PT Intelligent Sensing-IoT, Telkom University (2026)
- Internship Certificate — PT SAS Aero Sishan (2026)
- Electrical Circuits Teaching Assistant & Practicum Division Member — Instrumentation System Laboratory, Telkom University (2025)
- Electronics Teaching Assistant & Practicum Division Member — Instrumentation System Laboratory, Telkom University (2025)
- Certificate of Appreciation — Mentor Division, Kalibrasi 2024 Organizing Committee, HMTF (2025)
- Cadre Development Department Staff — HMTF Unitel, 2024/2025 Board Term (2025)
- Programming Laboratory Teaching Assistant & Rules and Discipline Committee Member — Basic Computing Laboratory, Telkom University (2024)
- Digital Learning Study Group — MBC Lab, Telkom University (2024)
- Using the Terminal/CMD for Development — Codepolitan (2024)
- Basic Intensive English Program — Language Center (LC), Kampung Inggris Pare; one-month course, average score 89/100 (2022)